

wanting to pursue higher returns without being able to short the market (to remove small caps' market exposures) or an investor could tilt to small caps because she wishes high returns but cannot lever. The unconstrained, investment-only reason for small caps, however, is not compelling.

3.3. VALUE FACTOR

Unlike size, the value premium is robust. Figure 7.6 graphs cumulated returns on the value strategy, *HML*. Value has produced gains for the last fifty years.²⁵ There are several notable periods where value has lost money in Figure 7.6, some extending over several years: the recession during the early 1990s, the roaring Internet bull market of the late 1990s, and there were large losses from value strategies in the financial crisis over 2007–2008. The risk of the value strategy is that although value outperforms over the long run, value stocks can underperform growth stocks during certain periods. It is in this sense that value is risky.

The benefits of value have been known since the 1930s. Graham and Dodd published a famous book, *Security Analysis*, in 1934, that serves as a guide to identifying firms with low prices relative to their fundamental value. Academics and practitioners proxy fundamental value today by various balance sheet variables or transformations thereof. Graham and Dodd were at Columbia Business School, where I teach, and a strong value-investing tradition continues at my institution today with the Heillbrunn Center for Graham & Dodd Investing. Modern academic research into the value effect began with Basu (1977), and the last few decades have seen an explosion of papers offering various explanations for the value premium. These explanations, like most of the finance literature, fall largely into two camps: the rational and the behavioral.

3.4. RATIONAL THEORIES OF THE VALUE PREMIUM

In the rational story of value, value stocks move together with other value stocks after controlling for market exposure (and in fact covary negatively with growth stocks). All value stocks, therefore, tend to do well together or they do badly together. Finding a value stock that doesn't move together with the pack is like finding the sourpuss not thrashing at a heavy metal concert. Just as the euphoria can't last, and some fans experience throbbing migraines the next day, value doesn't always earn high returns. Value is risky, and the riskiness is shared to a greater or lesser degree by all value stocks. Some value risk can be diversified by creating portfolios of stocks, but a large amount of value movements cannot be diversified away. (In fact, Fama and French exploit this *common covariation* in constructing the *HML* factor.) In the context of the APT, since not all risk can be

²⁵ Interestingly, a few authors including Ang and Chen (2007) show that value did not exist in the first half of the twentieth century.